

SIOP Machine Learning Competition 2026

Team Hungry Llama

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ForsMarsh

Team Members



Shane Halder



Jen Gibson

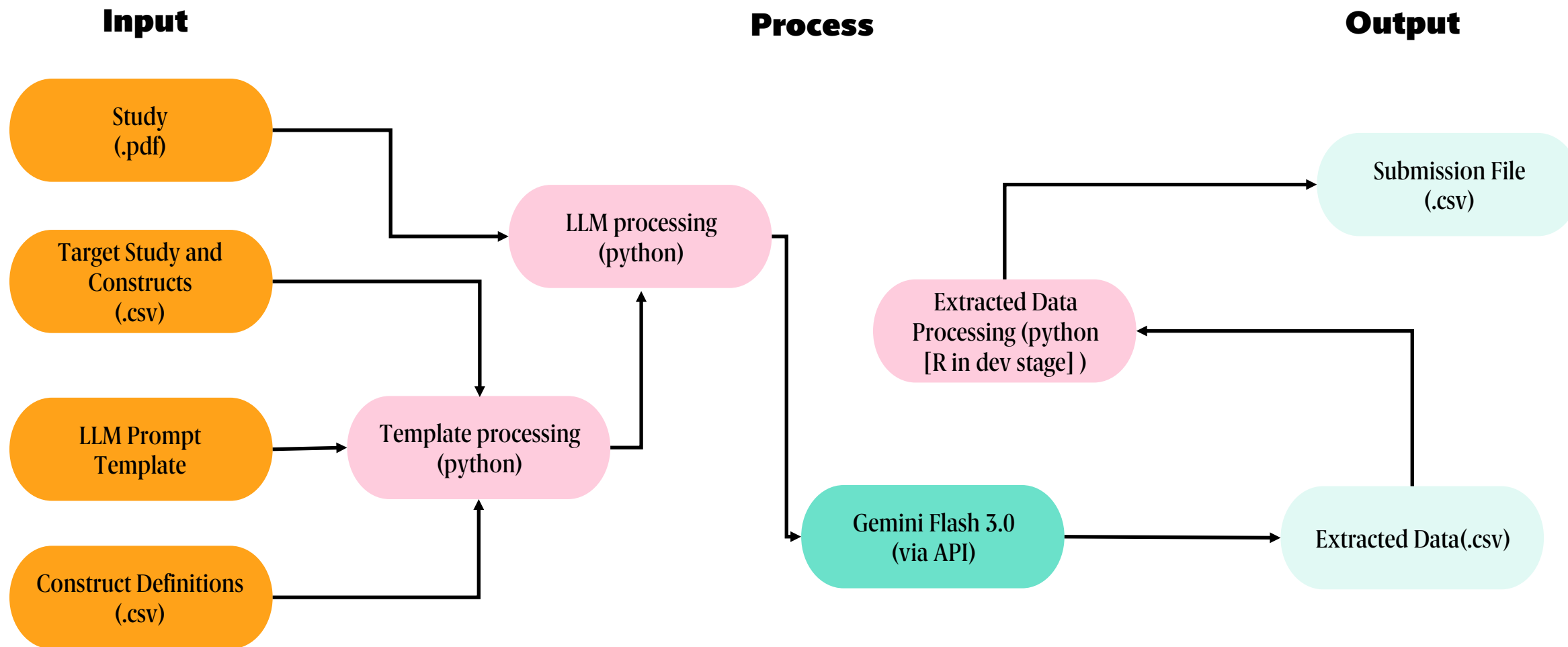


Joseph Luchman



Ron Vega

Pipeline Overview



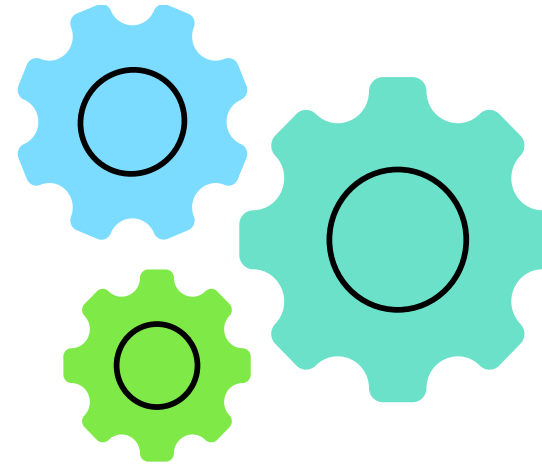
LLM specification

Solution used Gemini Flash 3.0

- Good balance between intelligence and cost efficiency
- Temperature = default (1.0)
- Thinking level = high

Other LLMs were tested and performed not as well

- Gemini 3.1
- OpenAI GPT-5, 5.2, 5.4
- Considered an ensemble between Gemini and GPT, but Gemini routinely surpassed GPT



LLM Prompt

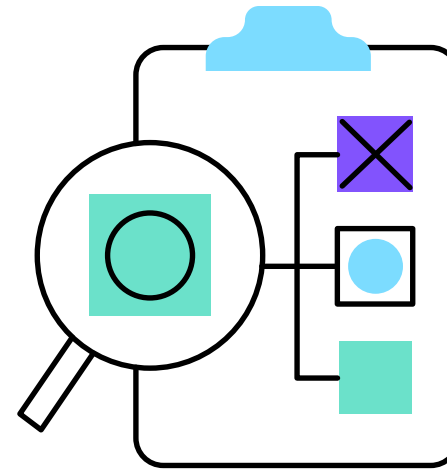
Prompt template described the task and defined the specifications of the data extraction

- Exclude effects where both variables reflect a single construct
- Exclude cross-lagged pairs (Time 1 X -> Time 2 Y)
- Extract effect sizes as-is including a negative sign if it exists
- Tables containing effect sizes may be rotated

Python code was used to pipe in construct definitions into prompt

- Prompt template and construct definitions were refined through SME review and consideration of outlying study extraction details

Extracted many different effects, most of which were not used



Structured Output (JSON) Feature

Metadata

- Title
- Author
- Total pages

Effects

- Pearson's r
- Logit and probit coefficients
- Odds Ratios
- Linear Regression
- ANOVA
- Cohen's d
- T-test
- Chi-square
- Other

Effect

- Page number
- X/y variable label
- X/y variable description
- X/y variable valence
- X/y variable reversed scored
- n (sample size)
- coefficient

Aggregation of Effects

During dev phase, we originally converted all effect sizes to r and calculated sample size weighted averages of effects

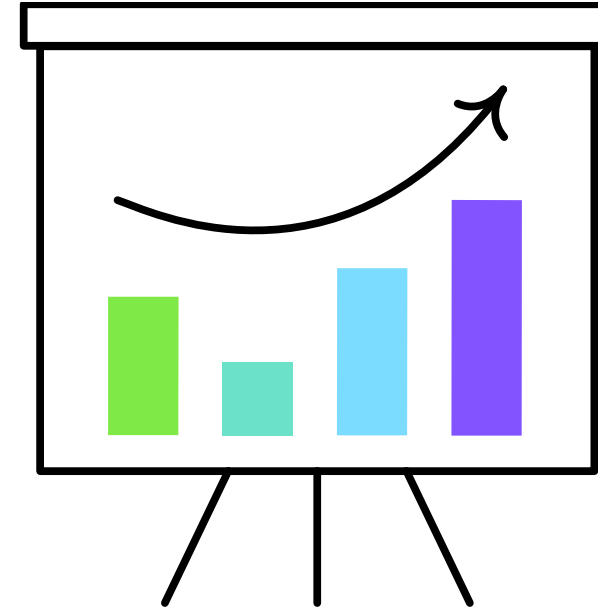
- Almost every dev stage submission resulted in worse scores than submitting only extracted r and taking a simple average

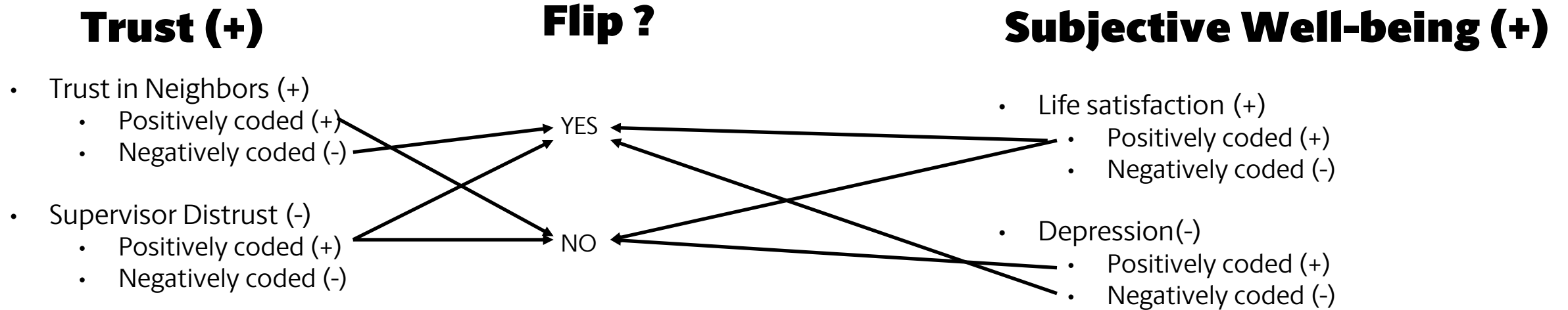
In the dev phase, we also needed to address what to do with studies that had no effect size extracted

- “Crud Factor” (Orben & Lakens, 1990)
- Previous meta-analyses imputation
- Test phase at least 1 effect was found for all studies

One aggregation improvement that resulted in performance gain was an increased focus on the valence of the effect sizes

- Let LLM decide
- Absolute value of effect
- Hard coding of effect directionality





Valence for predictor and outcome

Operationalization outcome for predictor and outcome

Coding directionality for predictor and outcome

$2^6 = 64$ different conditions!

Takeaway Thoughts

- LLMs can be useful for extraction tasks, but researchers should consider ultimate use
 - For meta-analyses, there is intrinsic value to the researchers to go through the articles and develop an in depth understanding of the constructs
 - Better understanding of constructs => Better LLM prompts
 - For getting a parameter estimate for a simulation study, Bayesian prior, or synthetic audience, an LLM extraction is a time saving option

https://colab.research.google.com/drive/1jIC1Seb-bnJvUHOUB_SfFHpxvClOFhlq?usp=sharing

Thank you.